A03 — Analysis of AI Use Cases in Healthcare & Agriculture

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Healthcare

Healthcare is one of the largest and most complex industries in the world. It covers hospitals, clinics, pharmaceutical research, insurance systems, and medical devices, all working together to prevent, diagnose, and treat illness. In the United States, healthcare makes up roughly 18% of the country's economy and employs over 22 million people. Around the world, it serves billions of patients in very different settings, from advanced hospitals in major cities to small rural clinics with limited resources.

The industry is under real pressure right now. The global population is getting older, which means more people need long-term care for conditions like diabetes and heart disease. There are not enough doctors in many parts of the world to meet that demand. Developing a new drug takes an average of 10 to 15 years and costs over one billion dollars, which puts many treatments out of reach. The American Medical Association (AMA) prefers to call AI in medicine *augmented intelligence* rather than *artificial intelligence*, because the goal is for these tools to support doctors, not replace them (American Medical Association and Manatt Health). Even so, while 65% of physicians surveyed by the AMA see clear benefits in AI tools, 70% are either more concerned than excited or equally both (American Medical Association and Manatt Health).

AI-driven diagnostics are the most established use of AI in healthcare today. As of late 2023, the FDA has approved 692 AI and machine learning medical devices, with the majority in radiology, cardiology, and neurology (American Medical Association and Manatt Health). These systems use computer vision to find problems in X-rays, MRIs, and scans, often matching or outperforming human specialists in specific tasks. One major milestone was the FDA's 2018 approval of the first autonomous AI system for detecting diabetic retinopathy. In a study of 900 patients, it achieved a sensitivity of 87.2% and specificity of 90.7%, beating the required clinical targets, meaning a regular primary care doctor can now screen for a leading cause of blindness without a specialist referral (American Medical Association and Manatt Health). Google DeepMind built a similar system for eye disease detection, and at the University of Florida, researchers worked with NVIDIA to develop GatorTron, an AI model trained on clinical data that helps doctors interpret complex patient histories (Deepgram).

Drug discovery is another area where AI is making a real difference. Moderna used AI to speed up the design of its mRNA COVID-19 vaccine, compressing a process that normally takes years (TechTimes Editorial). Insilico Medicine created a drug called ISM001-055 that became the first AI-designed drug to pass Phase IIa clinical trials, reducing the time from project start to clinical candidate by over 60% (Crescendo AI). That result caught the attention of Eli Lilly, which signed a deal worth up to $2.75 billion for rights to develop Insilico's AI-discovered drug candidates (Crescendo AI). DeepMind's AlphaFold, which predicts the 3D structure of proteins, won a Nobel Prize in Chemistry and is now a standard tool for researchers studying how molecules interact in the body (TechTimes Editorial).

AI is also changing how doctors work day to day. Robotic surgical systems now use AI to improve precision and guide surgeons in real time, leading to faster patient recovery in many procedures. Microsoft's DAX Copilot, which listens to doctor-patient conversations and automatically writes up clinical notes, has been adopted by more than 150 health systems (Deepgram). A study by the American Academy of Family Physicians found that doctors using AI documentation assistants cut their documentation time by 50 to 72% and saved over 3.3 hours per week (American Medical Association and Manatt Health). AI-powered chatbots and virtual counselors are also expanding access to mental health support for patients who cannot easily see a provider in person.

Bias in AI diagnostic tools is a real and documented problem, not just a theoretical one. Because these systems learn from historical data, they pick up on existing inequities in that data. The AMA reported one case where a care management algorithm consistently recommended white patients over Black patients for extra support. The reason was that it used past healthcare costs as a stand-in for health needs. Since Black patients had historically faced more barriers to care and had lower recorded costs, the algorithm wrongly assumed they were healthier (American Medical Association and Manatt Health). Another study found that pulse oximeters are less accurate on patients with darker skin, causing some ICU patients to receive less oxygen than they actually needed (American Medical Association and Manatt Health).

Beyond bias, two other ethical issues stand out. The first is explainability. Most top-performing AI models are too complex for anyone to trace exactly how they reach a conclusion. Doctors are legally responsible for their decisions, yet they often cannot verify the AI's reasoning before deciding whether to trust it, and the liability question that creates is still legally unresolved (American Medical Association and Manatt Health). Related to this are hallucinations, where AI systems produce responses that sound credible but are factually wrong. In a clinical setting, a hallucinated drug interaction or incorrect patient history could seriously harm someone (American Medical Association and Manatt Health). The second issue is privacy. HIPAA, the main law protecting patient data in the US, was written before AI became part of everyday medicine and does not fully address how AI systems today collect, store, or potentially re-identify patient information (American Medical Association and Manatt Health).

Looking ahead 5 to 10 years, the biggest shift in healthcare AI will likely be a move from treating illness to predicting and preventing it. Wearables already track heart rate, blood sugar, and sleep. In the near future, those signals will feed into AI models that warn patients and doctors before a health crisis happens (University of Florida Health). Drug development timelines are expected to drop from 10 to 15 years to around 3 to 5 years for many drug types, with the first fully AI-designed drugs expected to receive FDA approval in the coming years (Deepgram). The class materials also pointed to multi-omics data integration, AI in tissue engineering, quantum computing for biological simulations, and brain-computer interfaces as longer-term directions (Ganti). The key workforce risk is that healthcare systems that do not retrain their staff alongside these tools could end up offering two very different standards of care depending on where a patient seeks treatment.

Agriculture

Agriculture is the foundation of human civilization. It produces the food, fiber, and raw materials that every other industry and every person depends on. Globally, it employs over one billion people and accounts for about 4% of the world's economy, though in developing countries that share is much higher. It covers everything from row crops and livestock to aquaculture, horticulture, and the entire supply chain that gets food from a field to a grocery store.

The pressure on agriculture right now is intense. Climate change is disrupting growing seasons and making extreme weather more frequent. The world population is close to 8.2 billion and is expected to reach 10 billion by 2050, meaning farmers will need to produce about 50% more food on the same amount of land or less (Intellias). Young people are leaving rural areas for cities, creating labor shortages on farms. Meanwhile, the cost of fertilizer, water, and fuel keeps rising while profit margins for most farmers stay tight. The AI market in agriculture is projected to grow from $1.7 billion in 2023 to $4.7 billion by 2028, which shows just how seriously the industry is taking this challenge (Intellias).

Precision farming is one of the most impactful uses of AI in agriculture. Instead of applying water, fertilizer, or pesticide uniformly across an entire field, AI systems analyze real-time data to treat each section of the field based on what it actually needs. Blue River Technology, which John Deere developed into its See and Spray platform, uses computer vision to identify individual weeds and spray herbicide only on those specific plants, reducing total chemical use by up to 90% in some cases (Intellias). Prospera uses AI to monitor crop health across entire operations continuously, and Bayer's Climate FieldView platform analyzes satellite imagery and sensor data to spot crop stress early and suggest targeted fixes (Farmonaut). AI-connected irrigation systems can also decide in real time how much water specific crops need, cutting waste while protecting yields (Intellias).

Autonomous equipment and AI-powered disease detection are both in wide use today. John Deere's self-driving tractor uses AI, GPS, and computer vision to plant, cultivate, and harvest with minimal human operator input (Intellias). Computer vision tools can detect apple black rot with over 90% accuracy and identify pest insects at the same level (Intellias). Drone fleets with multispectral and thermal cameras can scan an entire field in hours, catching disease or drought stress before it is even visible to the human eye. CattleEye uses drones and computer vision to monitor cattle health remotely, spotting unusual behavior and tracking how diet and environment affect individual animals (Intellias). On the data side, AI models can now forecast harvest windows and quality grades before a crop goes in the ground, and blockchain-based tracking systems can trace a contamination problem back to its source in minutes rather than days (The Data Community).

Data ownership is one of the most pressing ethical concerns in AI-driven agriculture. When a farmer uses a platform like Climate FieldView, they generate detailed data about their land, soil, yield, and inputs. The companies running these platforms often keep the right to use that data to improve their own models, which they then sell back to farmers as upgraded services. Large agricultural businesses generate far more data than small family farms, so they get better and more tailored recommendations. The result is that the technology meant to help all farmers may end up giving the biggest advantage to those who already have the most (Intellias). On top of that, the Intellias report flagged the risk of cyberattacks targeting farming systems with the intent of disrupting food supplies, meaning the same digital infrastructure that makes smart farming possible creates real security vulnerabilities (Intellias).

The digital divide is another serious problem. Autonomous equipment, satellite subscriptions, and AI advisory tools are expensive, and most small-scale farmers, especially in developing countries, cannot afford them or do not have the infrastructure to use them (Ganti). This means AI could end up widening the gap between large industrial operations and smaller farms rather than helping everyone. There is also the labor question. Autonomous tractors, robotic harvesters, and AI-directed systems are replacing farm workers in communities that often have very few other job options and limited access to retraining programs, unlike displaced factory workers in urban areas who tend to have more support available (Intellias).

In the near future, the biggest change will be edge AI, meaning AI that runs directly on the tractor, drone, or sensor without needing a cloud connection. That makes these tools usable in remote areas where internet access is unreliable, which is where most farming actually happens (DLL Group). Over the next decade, AI-optimized vertical farming will become more common in cities, growing food indoors with AI controlling light, temperature, nutrients, and CO2 at the plant level, using far less water and producing much higher yields per square foot than outdoor farming. The class materials also pointed to AI combined with gene editing to develop climate-resilient crops, quantum computing for simulating complex ecosystems, and AI helping optimize sustainable protein sources like lab-grown meat and insect farming (Ganti). The main long-term risk is monoculture. When AI pushes farmers toward the highest-yield varieties, genetic diversity drops and the food system becomes more vulnerable to a single disease wiping out a major crop. The Irish Potato Famine is the clearest historical example of what happens when that goes wrong.

Reflection

What really caught me off guard between these two AI applications was healthcare. It's wild how much it has to juggle, from patients struggling to describe their symptoms accurately to dealing with biased data across different races and genders. I used to think healthcare AI would just be a specialized tool for a second opinion, not something acting as a primary guide for treatment. My own background involves working on a model designed to find people lost at sea—it was trained to recognize a lot of things but focused mostly on human detection. Seeing that same kind of vision technology being used in medicine is pretty incredible to me; it feels like a huge leap forward for humanity, especially when it comes to better cancer detection.

As an AI engineer, I feel a real sense of responsibility regarding the bias in LLMs. To me, this is one of the biggest risks since these models can end up racially profiling people or putting lives in danger based on sex or race. It goes hand-in-hand with hallucinations; if a single model is running everything, there's a high chance it'll misremember something and state it as absolute truth. That just gives people a false sense of security when the AI should really be saying it needs more data before making a call. I think the best way around this is using multi-model systems to minimize those hallucinations and split tasks up so the data stays organized and the output is more honest.

Healthcare AI feels like the priority right now, but in the long run, I think Agriculture AI is actually more urgent. We don't really have rules to manage the human population, and looking at places like India currently trying to count every citizen, it makes you wonder how much our food sources can actually handle before they collapse without AI stepping in. The world is growing fast, and it's hard to tell if the planet can keep up with us much longer.

Honestly, I'd be happy working in either field because both are about finding solutions for our future. But if I had to pick one, I'd choose the farming side of AI. I like the idea of making an impact that isn't just individual but truly global. With the world changing so quickly, we really need answers for the food shortages that are headed our way.

Works Cited

"AGRO: An Autonomous AI Rover for Precision Agriculture." *Arxiv*, 2025, arxiv.org/pdf/2505.01200.

American Medical Association and Manatt Health. *Future of Health: The Emerging Landscape of Augmented Intelligence in Health Care*. AMA, 2024, www.ama-assn.org/system/files/ama-ai-opportunities-challenges.pdf.

Crescendo AI. "The Latest AI News and Breakthroughs in Healthcare and Medical." *Crescendo AI*, 21 Apr. 2026, www.crescendo.ai/news/ai-in-healthcare-news.

The Data Community. "AI in Agriculture: From Precision Farming to Autonomous Food Systems." *The Data Community*, 5 Jan. 2026, thedatacommunity.org/2026/01/05/ai-in-agriculture-from-precision-farming-to-autonomous-food-systems/.

Deepgram. "Medical AI Models Transforming Healthcare: 2026 Guide." *Deepgram*, 24 Mar. 2026, deepgram.com/learn/top-medical-ai-models-2026.

DLL Group. "Innovative Agriculture: Key Trends for 2026." *DLL Group*, 2026, www.dllgroup.com/en/blogs/blogsoverview/innovative-agriculture-key-trends-for-2026.

Farmonaut. "Artificial Intelligence in Agriculture: 7 Farming Trends 2026." *Farmonaut*, 17 Jan. 2026, farmonaut.com/blogs/artificial-intelligence-in-agriculture-7-farming-trends-2026.

Ganti, Sridhar. "AI in Healthcare and Agriculture." *ITAI 2372: Artificial Intelligence Applications*, Houston Community College, 2024. Class Slides.

Intellias. "AI in Agriculture and Farming: Revolutionizing Crop Growth." *Intellias*, 4 May 2026, intellias.com/artificial-intelligence-in-agriculture/.

TechTimes Editorial. "Biggest AI Breakthroughs in Healthcare That Are Transforming Modern Medicine." *TechTimes*, 9 May 2026, www.techtimes.com/articles/316451/20260509/biggest-ai-breakthroughs-healthcare-that-are-transforming-modern-medicine.htm.

University of Florida Health. "7 Medical Sciences Trends Shaping Healthcare in 2026." *UF Health*, 7 Jan. 2026, distance.physiology.med.ufl.edu/about/articles/7-medical-sciences-trends-shaping-healthcare-in-2026/.